

Mitoprep Answer Key 5

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- Q1) **Hint:** The magnetic force between two current-carrying wires is given by, $\frac{F}{l} = \frac{\mu_0}{4\pi} \frac{i_1 i_2}{r}$.

Step 1: Find the dimensions of each quantity.

$$\text{Dimensions of electric permittivity, } [\epsilon_0] = \frac{[q_1 q_2]}{4\pi [F] [r^2]} = [M^{-1} L^{-3} T^4 A^2]$$

$$\text{Dimensions of magnetic flux, } [\phi] = [B][A] = \frac{[F]}{[i][L]} [A] = [ML^2 T^{-2} A^{-1}]$$

$$\text{Dimensions of self-inductance, } [L] = \frac{[V]}{[\frac{di}{dt}]} = [ML^2 T^{-2} A^{-2}]$$

$$\text{Dimensions of magnetic permeability, } [\mu_0] = \frac{[F][r]}{[i_1 i_2][l]} = [MLT^{-2} A^{-2}]$$

- Q2) **Explanation:**

Hint: Round up the second significant digit.

Explanation: To round the number 978.02 to two significant figures, we start by identifying the first two non-zero digits, which are 9 and 7. The digit following them is 8, which is greater than 5. As a result, we round the number up. Therefore, 978.02 rounded to two significant figures is 980 or 9.8×10^2 .

Hence, option (3) is the correct answer.

- Q3) **Explanation:**

Hint: $v = \frac{dx}{dt}$

Step 1: Find the initial acceleration.

(given $x = \alpha t^4 + \beta t^2 + \gamma t + \delta$.)

we know that $v = \frac{dx}{dt}$

$$\frac{dx}{dt} = v = 4\alpha t^3 + 2\beta t + \gamma + 0.$$

now initial velocity at time $t = 0$.

$$\Rightarrow v_i = \gamma$$

Step 2: Find the ratio of its initial velocity to its initial acceleration.

$$v = 4\alpha t^3 + 2\beta t + \gamma$$

we know that $a = \frac{dv}{dt}$

$$\frac{dv}{dt} = 12\alpha t^2 + 2\beta$$

now initial acceleration at time $t = 0$.

$$\Rightarrow a_i = 2\beta$$

Therefore the ratio of its initial velocity to its initial acceleration is given as: $\frac{v_i}{a_i} = \frac{\gamma}{2\beta}$

Hence, option (4) is the correct answer.

Q4) **Explanation:**

Hint: Distance = Area under ($v-t$) graph

Step: Find the distance travelled in the given time interval.

Distance travelled = Area under $v-t$ graph;

$$\text{Distance travelled} = 2 \times \frac{1}{2} \times 20 \times 5$$

$$\text{Distance travelled} = 100 \text{ m}$$

Therefore, the distance travelled in the given time interval is 100 m.

Hence, option (2) is the correct answer.

Q5) **Explanation:**

Hint: $T = mg$

Step: Find the accelerations of the blocks A , B and C .

Given: Block A (4 kg) is on a frictional surface and blocks B (2 kg) and C (2 kg) are vertically hanging. The system is initially at rest due to friction.

Then, the string connecting B and C is cut.

When the string is cut, block C is no longer connected to the system and falls freely under gravity. So, the acceleration is g downward.

After the string is cut, block B is still attached to block A with the remaining string.

The spring or string pulls block B upward due to tension. So, the acceleration of block B is upward and equal to g , as only the tension acts on it.

As there is no additional horizontal force acting on the block A , so the acceleration of the block A is zero.

Therefore, the accelerations of the blocks A , B and C are zero, g upward, g downward.

Hence, option (3) is the correct answer.

Q6) **Hint:** Recall conservation of energy

Step 1: Write initial and final kinetic energies

$$K_i = \frac{1}{2}mv^2$$

$$K_f = 2mv^2$$

Q7) **Explanation:**

Hint: Apply conservation of linear momentum.

Step: Find the velocity of A , after the collision along the original direction.

The initial momentum of the particle A is $2mu\hat{i}$.

The initial momentum of the particle B is $mu(-\hat{i})$.

The net momentum of the system along the horizontal direction is given by;

$$\Delta p = 2mu\hat{i} - mu\hat{i} = mu\hat{i}$$

The velocity of A , after the collision along the original direction is given by;

$$v_A = \frac{mu}{m} = u \text{ m/s}$$

Hence, option (1) is the correct answer.

Q8) Explanation:

Hint: $v_{\text{avg}} = \frac{\text{Total displacement}}{\text{Total time taken}}$

Step 1: Find the time taken by the particle to cover a semi-circle.

The time taken by the particle to complete the semi-circle is given by:

$$t = \frac{d}{v}$$

$$\Rightarrow t = \frac{\pi R}{v}$$

Step 2: Find the average velocity of the particle.

The displacement of the particle upon covering a semi-circle is $2R$.

The average velocity of the particle is given by;

$$v_{\text{avg}} = \frac{\text{Total displacement}}{\text{Total time taken}}$$

$$\Rightarrow v_{\text{avg}} = \frac{2R}{\frac{\pi R}{v}}$$

$$\Rightarrow v_{\text{avg}} = \frac{2v}{\pi}$$

Hence, option (4) is the correct answer.

Q9) Explanation:

Hint: $W = \int \vec{F} \cdot d\vec{x}$

Step: Find the work done by a given force.

The work done by a variable force is given by;

$$W = \int \vec{F} \cdot d\vec{x}$$

$$\Rightarrow W = \int \vec{F} \cdot d\vec{x} = \int_0^4 (2 + 3x) \hat{i} \cdot (dx) \hat{i}$$

$$\Rightarrow W = \left[2x + \frac{3}{2}x^2 \right]_0^4$$

$$\Rightarrow W = 32 \text{ J}$$

Therefore, work done by the force is 32 J.

Hence, option (2) is the correct answer.

Q10) Explanation:

Hint: $H = \frac{u^2 \sin^2 \theta}{2g}$

Step: Find the relation between θ_1 and θ_2 .

It is given that $H_1 > H_2$,

$$\Rightarrow \frac{u^2 \sin^2 \theta_1}{2g} > \frac{u^2 \sin^2 \theta_2}{2g}$$

$$\Rightarrow \sin^2 \theta_1 > \sin^2 \theta_2$$

$$\Rightarrow \theta_1 > \theta_2$$

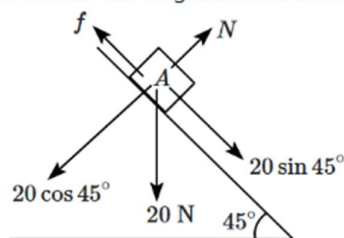
Hence, option (1) is the correct answer.

Q11) Explanation:

Hint: $f = \mu R$

Step 1: Draw the free-body diagram.

We can resolve the weight of the block and the components are shown in the figure below.



Step 2: Find the force of friction on the block.

$$N = 20 \cos(45^\circ) = \frac{20}{\sqrt{2}} = 10\sqrt{2} \text{ N}$$

The frictional force is given by;

$$\mu N = 2 \times 10\sqrt{2} \text{ N}$$

The friction force along the inclination is given by;

$$f = mg \sin(45^\circ) = \frac{20}{\sqrt{2}} = 10\sqrt{2} \text{ N}$$

Therefore, the net friction force on the 2 kg block is $10\sqrt{2} \text{ N}$.

Hence, option (3) is the correct answer.

Q12) Explanation:

Hint: $\Delta V = \text{area under } a - t \text{ graph.}$

Step 1:

Find area under $a - t$ curve.

$$\begin{aligned} \text{Area} &= 4 \times 4 \times \frac{1}{2} - 4 \times 1 \\ &= 4 \end{aligned}$$

Step 2:

Find change in velocity.

$$\begin{aligned} \Delta V &= \text{Area under } a - t \text{ curve} \\ &= 4 \end{aligned}$$

Step 3:

Find final velocity.

$$\begin{aligned} v_f - v_i &= \Delta V \\ v_f &= 4 + 3 \\ &= 7 \text{ m/s} \end{aligned}$$

Q13) Explanation:

Hint: Relative velocity is the slope of the graph of relative separation vs. time.

Step: Identify the correct match.

(A): It is given that the acceleration of both A and B is the same, therefore the relative velocity between the two particles must remain constant. If the relative velocity between the two particles is constant, then the graph of the relative separation between the particles over time should be a straight line. Among the given options, only the graph in option (B) matches these conditions.

(B): It is given that the acceleration of A is greater than that of B, while the initial velocity of B is higher than A's. Therefore, initially, the relative separation between the particles increases until their relative velocities become equal. At this point, the slope of the graph becomes zero. As time progresses, A's velocity will exceed B's, causing the relative separation between the particles to decrease, as depicted in the graph shown in option (IV).

(C): Initially, the velocities of A and B are the same, so the initial slope of the graph between relative separation and time (representing the relative velocity) will be zero. Since the acceleration of B is greater than that of A, B's velocity will eventually surpass A's, and the relative separation between them will keep increasing, as shown in the graph given in option (III).

(D): Initially, the velocity of B is greater than that of A, so the initial slope of the graph between relative separation and time (representing the relative velocity) will be positive and greater than zero. Since the acceleration of B is greater than A's, B's velocity will continue to increase compared to A's, and the relative separation will keep increasing, as shown in the graph given in option (I).

Therefore, the correct match is **A-II, B-IV, C-III, D-I**.

Hence, option **(3)** is the correct answer.

Q14) Explanation:

Hint: Distance travelled $= \int_{t_1}^{t_2} |v| dt$

Step 1:

Find velocity.

$$\begin{aligned} v &= \frac{ds}{dt} \\ &= 4 - 4t \end{aligned}$$

Step 2:

$$4 - 4t = 0$$

$$t = 1s$$

For $t < 1s, v > 0$

For $t > 1s, v < 0$

$$\begin{aligned} v_s &= |v| \\ &= 4 - 4t \quad t < 1s \\ &= 4t - 4 \quad t > 1s \end{aligned}$$

Step 3:

$$\begin{aligned} S &= \int v_s dt \\ &= \int_0^1 (4 - 4t) dt + \int_1^2 (4t - 4) dt \\ &= 4m \end{aligned}$$

Q15) **Explanation:**

Hint: $v^2 = u^2 + 2as$

Step 1: Find the minimum speed of the throw.

At the highest point, the velocity of the ball is zero.

By using the third equation of motion we get;

$$\Rightarrow v^2 = u_{min}^2 + 2as$$

$$\Rightarrow 0 = u_{min}^2 + 2 \times (-10) \times 15$$

$$\Rightarrow u_{min} = \sqrt{300} \text{ m/s} = 10\sqrt{3} \text{ m/s}$$

Step 2: Find the maximum speed of the throw.

The maximum speed of the throw is given by;

$$v^2 = u_{max}^2 + 2as$$

$$\Rightarrow 10^2 = u_{max}^2 - 2 \times 10 \times 15$$

$$\Rightarrow u_{max}^2 = 100 + 300 = 400$$

$$\Rightarrow u_{max} = \sqrt{400} = 20 \text{ m/s}$$

Therefore, the minimum and maximum speeds of the throw are $10\sqrt{3}$ m/s and 20 m/s.

Hence, option **(4)** is the correct answer.

Q16) **Explanation:**

Hint: Apply the work energy theorem.

Step: Find the value of x .

$$W_{\text{Resistive force}} = \Delta K.E$$

$$\Rightarrow -F \times 4 = \frac{1}{2}m\left(\frac{v^2}{9} - v^2\right) \dots (1)$$

$$\Rightarrow -F \times (4 + x) = \frac{1}{2}m(0 - v^2) \dots (2)$$

By dividing eqn (1) by (2) we get;

$$\frac{4}{4+x} = \frac{8}{9}$$

$$\Rightarrow 36 = 32 + 8x$$

$$\Rightarrow x = \frac{1}{2} = 0.5$$

Hence, option **(3)** is the correct answer.

Q17) **Explanation:**

Hint: x -component of velocity remains constant throughout the motion.

Step 1: Find the angle of projection, θ .

The velocity of the ball at maximum height is given by;

$$u \cos \theta = \frac{\sqrt{3}u}{2}$$

$$\Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = 30^\circ$$

Step 2: Find the time of flight of the projectile.

The time of flight of the projectile is given by;

$$T = \frac{2u \sin \theta}{g}$$

$$\Rightarrow T = \frac{2u \sin 30^\circ}{g}$$

$$\Rightarrow T = \frac{u}{g}$$

Hence, option **(3)** is the correct answer.

Q18) **Explanation:**

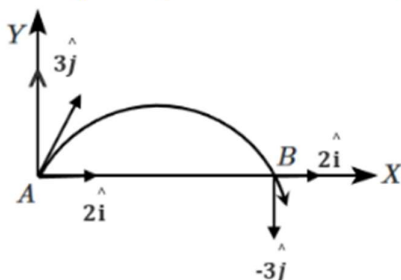
Hint: Use $\vec{v} = v_x \hat{i} + v_y \hat{j}$

Step: Find the velocity at the point B .

As the velocity of the projectile at the point A is given by;

$$2\hat{i} + 3\hat{j}$$

While coming at the point B , the x -component of the velocity will be in the positive x direction and the y -component of the velocity will be in the negative y direction as shown in the figure.



Therefore, the velocity at the point B is given by;

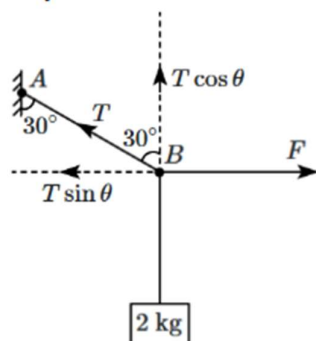
$$2\hat{i} - 3\hat{j}$$

Hence, option (2) is the correct answer.

Q19) **Explanation:**

Hint: Balance the forces along the vertical and horizontal directions.

Step 1: Draw the F.B.D.



Step 2: Find the value of the force (F).

$$T \sin \theta = F \dots (i)$$

$$T \cos \theta = mg \dots (ii)$$

From (i) and (ii)

$$F = mg \tan \theta$$

$$\Rightarrow 20 \left(\frac{1}{\sqrt{3}} \right) \text{ N}$$

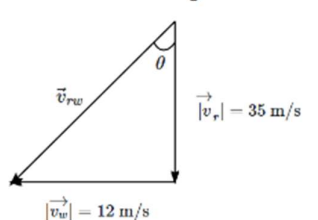
Hence, option (3) is the correct answer.

Q20) **Explanation:**

Hint: $\tan \theta = \frac{v_w}{v_r}$

Step: Find the direction in which the person should hold the umbrella.

If we draw a vector diagram for this resultant velocity of rain and wind, we get



If the umbrella makes an angle θ with the vertical, the direction of the resultant velocity of rain and wind should be the opposite direction of holding an umbrella.

$\tan \theta = \frac{v_w}{v_r}$

Substituting the known values we get,

$\tan \theta = \frac{12}{35}$

$\Rightarrow \theta = \tan^{-1}\left(\frac{12}{35}\right)$

Thus, as shown in the figure, a boy should hold the umbrella with the angle $\theta = \tan^{-1}\left(\frac{12}{35}\right)$

projecting himself from the rain.

Hence, option **(3)** is the correct answer.

Q21) **Explanation:**

Hint: When a bicycle moves with its brakes on, sliding friction is greater than rolling friction.

Explanation: It is difficult to move a bicycle along a road with its brakes on because sliding friction is greater than rolling friction. Brakes prevent the wheels from rolling and to move a bicycle with brakes on, it must be pushed like a body with no wheels so the bicycle skids on the road.

Therefore, both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

Hence, option (1) is the correct answer.

Q22) **Explanation:**

Hint: Velocity = $\frac{\text{displacement}}{\text{time}}$

Step: Find the time taken by Preeti to walk upon the moving escalator.

The velocity of Preeti with respect to the elevator, $v = \frac{d}{t_1}$

The velocity of the elevator with respect to the ground, $v = \frac{d}{t_2}$

Then the velocity of Preeti with respect to the ground is given by;

$v = v_1 + v_2$

$\Rightarrow \frac{d}{t} = \frac{d}{t_1} + \frac{d}{t_2}$

$\Rightarrow t = \frac{t_1 t_2}{t_1 + t_2}$

Here, t is the time taken by Preeti to walk upon the moving escalator.

Hence, option **(2)** is the correct answer.

Q23) Explanation:

Hint: Recall the dimensions of physical quantities.

Step: Find the dimensions of the length.

In terms of h , c , and G length can be expressed as,

$$L = (h)^p (c)^q (G)^r$$

Writing dimensions on both sides, we get:

$$\begin{aligned} [M^0 L T^0] &= [M L^2 T^{-1}]^p [L T^{-1}]^q [M^{-1} L^3 T^{-2}]^r \\ &= [M^{p-r} L^{2p+q+3r} T^{-p-q-2r}] \end{aligned}$$

On comparing the powers of M , L , and T on both sides, we get;

$$p - r = 0, \quad 2p + q + 3r = 1 \quad \text{and} \quad -p - q - 2r = 0$$

On solving, we get;

$$p = r = \frac{1}{2} \quad \text{and} \quad q = -\frac{3}{2}$$

Therefore, the dimensions of length are, $L = (h)^{1/2} (c)^{-3/2} (G)^{1/2} = \frac{\sqrt{hG}}{c^{3/2}}$

Hence, option (1) is the correct answer.

Q24) Explanation:

Hint: Use the concept of relative error.

Step 1: Find the expression for the density of the wire.

The density of the wire is given by;

$$\begin{aligned} \rho &= \frac{m}{V} = \frac{m}{AL} \\ \Rightarrow \rho &= \frac{m}{\pi r^2 L} \end{aligned}$$

Step 2: Find the maximum possible percentage error in the density of the wire.

$$\begin{aligned} \left(\frac{\Delta \rho}{\rho} \times 100 \right) &= \left(\frac{\Delta m}{m} + 2 \frac{\Delta r}{r} + \frac{\Delta L}{L} \right) \times 100 \\ \Rightarrow \left(\frac{\Delta \rho}{\rho} \times 100 \right) &= \left(\frac{0.002}{0.4} + 2 \times \frac{0.001}{0.3} + \frac{0.02}{5} \right) \times 100 = \left(\frac{1}{200} + \frac{2}{300} + \frac{2}{500} \right) \times 100 \\ \Rightarrow \left(\frac{\Delta \rho}{\rho} \times 100 \right) &= \left(\frac{1}{2} + \frac{2}{3} + \frac{2}{5} \right) \% \\ \Rightarrow \left(\frac{\Delta \rho}{\rho} \times 100 \right) &= (0.5 + 0.66 + 0.4) \% = 1.56 \% \\ \Rightarrow \left(\frac{\Delta \rho}{\rho} \times 100 \right) &\approx 1.6 \% \end{aligned}$$

Hence, option (4) is the correct answer.

Q25) Explanation:

Hint: Recall the relative error.

Step: Find the error in the determination of σ .

The relative error in the determination σ is given by;

$$\begin{aligned} \left(\frac{\Delta \sigma}{\sigma} \right) \times 100 &= \frac{\Delta E}{E} \times 100 + \frac{\Delta A}{A} \times 100 + 4 \frac{\Delta T}{T} \times 100 \\ \Rightarrow \left(\frac{\Delta \sigma}{\sigma} \right) \times 100 &= 1\% + 1\% + 4 \times 0.5\% = 4\% \end{aligned}$$

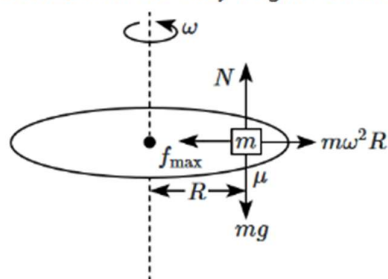
Therefore, the error in the determination of σ is 4%.

Hence, option (2) is the correct answer.

Q26) **Explanation:**

Hint: Centrifugal force = $m\omega^2 R$

Step 1: Draw the free-body diagram of the block.



Step 2: Find the maximum value of R .

$$f_{max} = \mu N$$

$$= \mu(mg)$$

$$f_{max} \geq mR\omega^2$$

$$\Rightarrow \mu mg \geq mR\omega^2$$

$$\Rightarrow R \leq \frac{\mu g}{\omega^2}$$

Hence, option (1) is the correct answer.

Q27) **Explanation:**

Hint: At point B particle will have only a horizontal velocity component.

Step: Find the velocity at the point B .

The horizontal component of the projectile remains constant, and at the highest point, the vertical component of the velocity is zero.

The velocity of the ball at the highest point is given by;

$$v_B = v \cos \theta$$

$$\Rightarrow v_B = 20 \cos 60^\circ$$

$$\Rightarrow v_B = 20 \times \frac{1}{2} = 10 \text{ ms}^{-1}$$

Hence, option (4) is the correct answer.

Q28) **Correct Option is A**

Q29) **Explanation:**

Hint: $v^2 - u^2 = 2as$

Step: Find the final kinetic energy of the lift.

Given: $u = 2 \text{ m/s}$, $a = 2 \text{ m/s}^2$, $s = 6 \text{ m}$

By using the third equation of motion we get;

$$v^2 - u^2 = 2as$$

By multiplying $\frac{m}{2}$ both sides we get;

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = 2as \times \frac{m}{2}$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mas$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{1}{2} \times 500 \times 2^2 + 500 \times 2 \times 6$$

$$\Rightarrow \frac{1}{2}mv^2 = 1000 + 6000 = 7000 \text{ J} = 7 \text{ kJ}$$

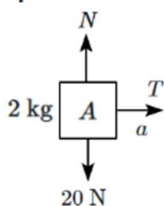
Therefore, the final kinetic energy of the lift is 7 kJ.

Hence, option (4) is the correct answer.

Q30) Explanation:

Hint: $F_{net} = ma$

Step 1: Draw the F. B. D. of both blocks.



Step 2: Find the acceleration of block A.

$$T = 2a \quad \dots (1)$$

$$20 - T = 2a \quad \dots (2)$$

From (1) & (2) we get,

$$\Rightarrow 20 - 2a = 2a$$

$$\Rightarrow 4a = 20$$

$$\Rightarrow a = 5\text{m/s}^2 \quad \dots (3)$$

Step 3: Find the tension in the connecting string.

Using (1) & (3) we get,

$$\Rightarrow T = 10\text{ N}$$

Hence, option (3) is the correct answer.

Q31) Explanation:

Hint: The ball is undergoing uniform accelerated motion.

Step: Find the average velocity of the ball.

Consider downward quantities as positive.

The ball is undergoing uniformly accelerated motion, so the average velocity is given by:

$$\begin{aligned} \vec{v}_{\text{avg}} &= \frac{\vec{v}_i + \vec{v}_f}{2} \\ &= \frac{(-v_1) + (v_2)}{2} \\ &= \frac{v_2 - v_1}{2} \end{aligned}$$

Hence, option (1) is the correct answer.

Q32) Explanation:

Hint: Acceleration is defined as the change of velocity divided by the time interval.

Step: Analyse each option one by one.

1. Any motion in a straight line: There are two motions possible in this case. One with constant velocity motion which can have zero acceleration however the other motion is of changing speed (non-uniform motion), which involves acceleration. Thus, not all straight-line motions have zero acceleration.

2. Projectile motion: This always involves acceleration due to gravity acting on the projectile. Even if the horizontal component of the velocity is constant, the vertical component changes due to gravity, resulting in a non-zero acceleration.

3. Any motion in a circle: Circular motion involves centripetal acceleration, which is always directed toward the center of the circle. Hence, any object moving in a circular path has a non-zero acceleration.

Therefore, none of the motions above guarantees zero acceleration.

Hence, option (4) is the correct answer.

Q33) **Explanation:**

Hint: $x = v \times t$

Step 1: Find the time taken to cross the river.

The time taken to cross the river is given by;

$$t = \frac{\text{Distance across the river}}{\text{Speed perpendicular to the river}}$$

$$\Rightarrow t = \frac{1 \text{ km}}{2 \text{ km/h}}$$

$$\Rightarrow t = 0.5 \text{ h}$$

Step 2: Find the distance that drifted downstream.

While the boat crosses the river, it is carried downstream by the river's flow. The distance drifted downstream (x) is given by; $x = v_r \times t$

$$\Rightarrow x = 3 \text{ km/h} \times 0.5 \text{ h}$$

$$\Rightarrow x = 1.5 \text{ km}$$

Hence, option **(3)** is the correct answer.

Q34) **Explanation:**

Hint: $a = \sqrt{a_x^2 + a_y^2}$

Step 1: Find the forces acting on the string before the string was cut.

Given: The system is stationary, meaning the applied force F is balanced.

The length of the string is 1 m, the angle made by the string with the vertical $\theta = 45^\circ$.

Initially, the vertical component of the tension T_y balances the weight of the bob i.e.,

$$T \cos 45^\circ = 20$$

$$\Rightarrow T = 20\sqrt{2} \text{ N}$$

The horizontal component of the tension balances the force F i.e., $T \sin 45^\circ = F$

$$\Rightarrow F = \frac{T}{\sqrt{2}} = \frac{20\sqrt{2}}{\sqrt{2}} = 20 \text{ N}$$

Step 2: Find the net acceleration of the bob after cutting the string.

The horizontal component of the acceleration is given by;

$$a_x = \frac{20}{2} = 10 \text{ m/s}^2$$

The vertical component of the acceleration is given by acceleration due to gravity;

$$a_y = 10 \text{ m/s}^2$$

The net acceleration of the bob is given by;

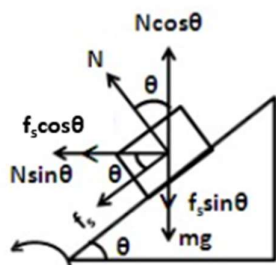
$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{10^2 + 10^2} = 10\sqrt{2} \text{ m/s}^2$$

$$\Rightarrow a = 10\sqrt{2} \text{ m/s}^2$$

Hence, option **(3)** is the correct answer.

Q35) **Step 1:** Draw FBD.

The following figure describes the situation mentioned in the question:



Step 2: Balance the vertical and the horizontal forces.

For vertical equilibrium,

$$N \cos \theta = mg + f_s \sin \theta$$

$$\Rightarrow mg = N \cos \theta - f_s \sin \theta \quad \dots (i)$$

For horizontal equilibrium,

$$N \sin \theta + f_s \cos \theta = \frac{mv^2}{R} \quad \dots (ii)$$

Step 3: Find the maximum safe speed.

Divides eq. (i) and (ii), we get,

$$\frac{v^2}{Rg} = \frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} \quad [f_s = \mu_s N]$$

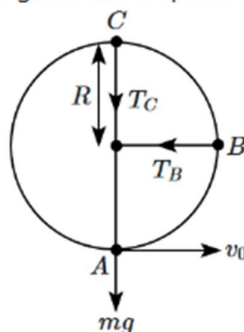
$$\Rightarrow v = \sqrt{Rg \left(\frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} \right)}$$

$$\Rightarrow v = \sqrt{Rg \left(\frac{\tan \theta + \mu_s}{1 - \mu_s \tan \theta} \right)}$$

Q36) Correct Option Is B

Q37) **Step 1:** Find the initial and final energy of the particle.

As given in the question, it can be represented in the form of the figure as follows:



Amount of energy at point A = $\frac{1}{2}mv_0^2 \dots (i)$

Amount of energy at point C = A = $\frac{1}{2}mv_c^2 + mg \times 2R \dots (ii)$

Step 2: Balance the force at the topmost point and apply the condition of minimum velocity.

Applying Newton's second law now at point C

$$T_c + mg = \frac{mv_c^2}{R}$$

Now for completing the loop $T_c \geq 0$

Therefore, the equation becomes :

$$mg = \frac{mv_c^2}{R} \Rightarrow v_c = \sqrt{Rg}$$

Step 3: Conserve the energy between the bottom and the topmost points.

Applying the law of conservation of energy, from eqn (i) and (ii),

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_c^2 + 2mgR$$

Putting equation (iii) in this equation we get,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mgR + 2mgR$$

$$v_0^2 = gR + 4gR$$

$$\Rightarrow v_0^2 = 5gR \Rightarrow v_0 = \sqrt{5gR}$$

Hence, option (3) is the correct answer.

Q38) **Explanation:**

Hint: Recall the dimensions of the given quantities.

Step: Find the quantities which have the same dimensions as those of solid angles.

The expression of the solid angle is given by; $\Omega = \frac{\text{Area}}{r^2} = \frac{[L^2]}{[L^2]} = [M^0L^0T^0]$

Therefore, the solid angle is a dimensionless quantity.

The dimensions of strain are given by; $\frac{\Delta L}{L} = [M^0L^0T^0]$

The dimensions of the angle are given by; $\theta = \frac{l}{r} = [M^0L^0T^0]$

Therefore, strain and angle are both dimensionless quantities.

Hence, option (4) is the correct answer.

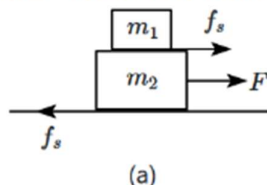
Q39) **Explanation:**

Hint: Kinetic friction acts opposite to the direction of motion.

Step: Analyse each statement one by one.

Static friction is what allows an object to start moving when a force is applied. It can accelerate an object from rest.

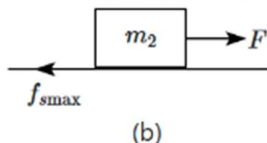
Let us consider the situation shown in the figure below (a),



For the moment of the above block, the static friction supports the block's motion. therefore, static friction can accelerate an object from rest.

Kinetic friction acts when an object is already sliding and cannot prevent slipping; it works to slow down the sliding motion. So, the second statement is false.

Kinetic friction acts on a moving object, opposing its motion, and can bring it to a stop. This statement is correct as shown in figure below (b),



Therefore, statement I and statement III are correct.

Hence option **(1)** is the correct answer.

Q40) **Explanation:**

Hint: $F = -\frac{dU}{dr}$

Step: Find the distance of the particle from centre of the field.

Given: The potential energy of the particle in a force field is; $U = \frac{A}{r^2} - \frac{B}{r}$

For stable equilibrium; $F = -\frac{dU}{dr} = 0$

$$\Rightarrow \frac{dU}{dr} = -\frac{2A}{r^3} + \frac{B}{r^2} = 0$$

$$\Rightarrow B = \frac{2A}{r}$$

The distance of the particle from centre of the field is given by;

$$r = \frac{2A}{B}$$

Hence, option **(3)** is the correct answer.

Q41) **Explanation:**

Hint: Check the dimensions of L.H.S. and R.H.S.

Step: Use formula for the time period of the simple pendulum

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T^2 = 4\pi^2\left(\frac{l}{g}\right) \dots (1)$$

The expression of the time period given in the question is; $T^2 = l/g \dots (2)$

From equations (1) and (2) we observed that it is dimensionally correct but not numerically.

Therefore, this equation is dimensionally correct but not numerically.

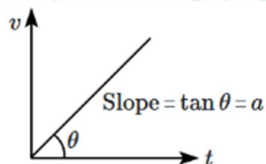
Hence, option (3) is the correct answer.

Q42) **Explanation:**

Hint: The graph is a straight line with a positive slope.

Step: Find the correct graph for the motion of the car.

The slope of the v - t graph gives the acceleration as shown in the figure below.



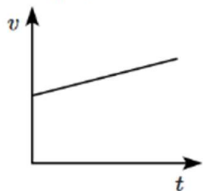
(1) In the first graph, the slope is positive hence acceleration is positive.

(2) In the second graph, the velocity decreases with time hence acceleration is negative.

(3) In the third option, the velocity decreases with time hence acceleration is negative.

(4) In the fourth option, the velocity remains constant with time hence acceleration is zero.

Therefore, the graph for the uniform positive acceleration is shown in the figure below;



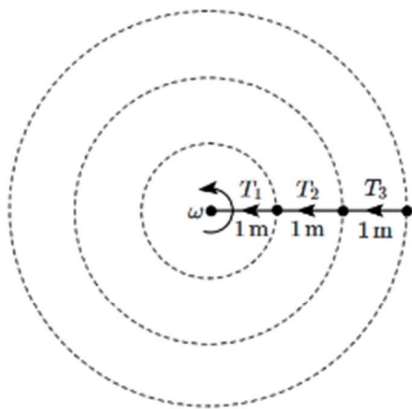
Hence, option (1) is the correct answer.

Q43) **Explanation:**

Hint: $T = \frac{mv^2}{r} = mr\omega^2$

Step: Find the values of tensions applied to the masses.

The tension applied on each ball is shown in the figure below;



Tension T_1 is providing the centripetal force to all three balls i.e.,

$$T_1 = m_1 r_1 \omega^2 + m_2 r_2 \omega^2 + m_3 r_3 \omega^2 = 1 \times 1 \times 16 + 1 \times 2 \times 16 + 1 \times 3 \times 16$$

$$\Rightarrow T_1 = 96 \text{ N}$$

Tension T_2 is providing the centripetal force to the last two balls i.e.,

$$T_2 = m_2 r_2 \omega^2 + m_3 r_3 \omega^2 = 1 \times 2 \times 16 + 1 \times 3 \times 16$$

$$\Rightarrow T_2 = 80 \text{ N}$$

Tension T_3 is providing the centripetal force to the last ball i.e.,

$$T_3 = m_3 r_3 \omega^2 = 1 \times 3 \times 16 = 48 \text{ N}$$

Therefore, the values of tension T_1 , T_2 and T_3 are 96 N, 80 N and 48 N respectively.

Hence, option (3) is the correct answer.

Q44) **Explanation:**

Hint: Apply the principle of homogeneity.

Step: Find the dimensions of a , b and c .

The velocity of the particle is given by; $v = at + \frac{b}{t+c}$

By using the principle of homogeneity we get;

$$[v] = [LT^{-1}] = [a][t] = \frac{[b]}{[t+c]} = \frac{[b]}{[T]}$$

$$\Rightarrow [LT^{-1}] = [a][T] \Rightarrow [a] = [LT^{-2}]$$

$$\Rightarrow [c] = [T]$$

$$\Rightarrow [LT^{-1}] = \frac{[b]}{[T]} \Rightarrow [b] = [L]$$

Therefore, the dimensions of a , b and c are $[LT^{-2}]$, $[L]$, $[T]$ respectively.

Hence, option (1) is the correct answer.

Q45) **Explanation:**

Hint: $p = Fv$

Step: Find the change in acceleration.

As power is given by, $p = F \cdot v = \text{constant}$

As the velocity is doubled, the force will be halved and hence the acceleration as the mass of the body remains the same and the acceleration is directly proportional to the force.

Therefore, both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.

Hence, option **(3)** is the correct answer.

Q46) **Explanation:**

Hint: A species with one electron exhibits the same spectrum as hydrogen.

Q47) **Step 1: Determine the moles of each component.**

We are given the mass of each component and their molar masses. We will calculate the moles of each substance.

Potassium (K): Mass = 0.39 g Molar mass of K = 39 g/mol Moles of K = $\frac{0.39}{39} = 0.01 \text{ mol}$

Aluminum (Al): Mass = 0.27 g Molar mass of Al = 27 g/mol Moles of Al = $\frac{0.27}{27} = 0.01 \text{ mol}$

Sulfate radical (SO₄): Mass = 1.92 g Molar mass of SO₄ = 96 g/mol Moles of SO₄ = $\frac{1.92}{96} = 0.02 \text{ mol}$

Water (H₂O): Mass = 2.16 g Molar mass of H₂O = 18 g/mol Moles of H₂O = $\frac{2.16}{18} = 0.12 \text{ mol}$

Step 2: Derive the formula from mole ratios.

From the mole calculations, we can see the following approximate ratios:

Moles of K = 0.01 mol

Moles of Al = 0.01 mol

Moles of SO₄ = 0.02 mol

Moles of H₂O = 0.12 mol

This indicates a 1:1:2:12 ratio

which corresponds to the formula **KAl(SO₄)₂ · 12H₂O**

Step 3: Verify molar mass of above Empirical Formula.

Total molar mass = 39 + 27 + 192 + 216 = 474 g/mol

However, since the molar mass given in the problem is **948 g/mol**, this suggests the formula should be doubled.

(KAl(SO₄)₂ · 12H₂O)₂ which when rearranged it will be **K₂SO₄ · Al₂(SO₄)₃ · 24H₂O**

Hence Option 3 will be right answer.

Q48) **Explanation:**

Hint: The combining atomic orbitals must have the same symmetry about the molecular axis.

1. The π^* antibonding MO has a node between the nuclei. It is a correct statement. The node is a space where the electron density is zero.

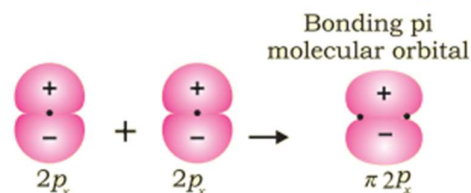
2. In the formation of a bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other. It is a correct statement. The formation of molecular orbitals can be understood in terms of the constructive or destructive interference of the electron waves of the combining atoms.

In the formation of a bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other due to constructive interference.

3. Molecular orbitals obtained from $2P_x$ and $2P_y$ orbitals are symmetrical around the bond axis. It is an incorrect statement.

The combining atomic orbitals must have the same symmetry about the molecular axis. For example, $2p_z$ orbital of one atom can combine with $2p_z$ orbital of the other atom but not with the $2p_x$ or $2p_y$ orbitals because of their different symmetries.

4. A π -bonding molecular orbital has a larger electron density above and below the internuclear axis. It is a correct statement. The representation is as follows:



Thus, option third is the correct answer.

Q49) **Explanation:**

Hint: Resonance structures are sets of Lewis structures that describe the delocalization of electrons in a polyatomic ion or a molecule.

Explanation:

The chemical formula of nitrogen(I)oxide is N_2O . The N_2O molecule has a total number of 18 valence electrons - 6 from nitrogen and 6 from each oxygen atom.

Three resonance structures of N_2O molecule are as follows:



Q50) **Explanation:**

Hint: Gay-Lussac's Law of Gaseous Volumes

Explanation:

Dalton theory could explain the laws of chemical combination. However, it could not explain the law of gaseous volumes. The law of gaseous volumes can only be explained with the help of ideal gas equation and mole-mole analysis .

In SO_2 and SO_3 the ratio of the masses of oxygen which combine with a fixed mass of sulphur is 2:3. This is an example of the law of multiple proportion. CO_2 and CO also follows law of multiple proportions.

Hence, both the statements are incorrect. So, option 4 is the correct answer.

Q51) Correct Option is A

Q52) **Hint:** Cation has more effective nuclear charge than anion.

In case of isoelectronic species, effective nuclear charge increases with increase in atomic number Z (with increase in protons). This results in decrease in size and hence, ionic radius of $\text{Al}^{3+} < \text{Na}^+$.

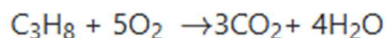
Q53) **Explanation:**

n = energy level and

l = subshell.

When $n = 4$ and $l = 3$, the subshell is $4f$. There are 7 orbitals on the f subshell. According to the Pauli exclusion principle, each of these orbital can contain a maximum of two electrons. therefore the maximum number of electrons in $4f = 7 \times 2 = 14$

Q54) **Explanation:**



$$22.4\text{L} \quad 5 \times 22.4\text{L}$$

For the combustion of 22.4L propane, oxygen required = $5 \times 22.4\text{L}$

For the combustion of 1L of propane oxygen required = $5 \times 22.4 / 22.4\text{L}$

$$= 5\text{L}$$

Q55) Explanation:

HINT: Isoelectronic species have the same number of electrons.

Explanation:

Ions	Electronic configuration	Number of electrons
Mn ⁺²	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁵	23
Fe ⁺³	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁵	23
Fe ⁺²	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁶	24
O ⁻²	1s ² 2s ² 2p ⁶	10
F ⁻	1s ² 2s ² 2p ⁶	10
Na ⁺	1s ² 2s ² 2p ⁶	10
Mg ⁺²	1s ² 2s ² 2p ⁶	10

Fe²⁺ has 24 electrons. Mn²⁺ has 23, hence they are not an iso-electronic pair.

All other pairs are isoelectronic.

Q56) Explanation:

$$\text{Hint: } \text{molality} = \frac{\text{number of moles of solute}}{\text{amount of solvent in kg}}$$

Step 1:

Calculate the amount of solvent in the solution as follows:

$$\text{density} = \frac{\text{mass of solution}}{\text{volume of solution}}$$

The molarity of the solution is 1 M. It indicates that 1 mole of solute is present in 1000 mL of solution or 1 L.

$$1.25 = \frac{\text{mass of solution}}{1000}$$

$$\text{mass of solution} = 1250 \text{ g}$$

$$\begin{aligned} \text{The amount of solute} &= \text{number of moles} \times \text{molar mass of a compound X} \\ &= 1 \times 85 \\ &= 85 \text{ g} \end{aligned}$$

$$\begin{aligned} \text{Amount of solvent} &= 1250 - 85 \\ &= 1165 \text{ g or } 1.165 \text{ kg} \end{aligned}$$

Step 2:

Calculate the molality as follows:

$$\begin{aligned} \text{molality} &= \frac{1}{1.165} \\ &= 0.858 \text{ m} \end{aligned}$$

Q57) **Explanation:**

Hint: Isoelectronic species contain the same number of electrons

Q58) In group 17, electronegativity, IE , and oxidizing power decrease from top to bottom. Hence, all these properties increase from iodine to fluorine.

Let's analyze which properties of halogens increase as we move from iodine to fluorine:

(a) **Bond length:** As we move from iodine to fluorine, the size of the atoms decreases, resulting in a decrease in bond length. So, this property does not increase from iodine to fluorine.

(b) **Electronegativity:** Electronegativity increases as we move up the group from iodine to fluorine. So, this property does increase from iodine to fluorine.

(c) **Ionization energy:** Ionization energy also increases as we move up the group from iodine to fluorine, due to the decrease in atomic size and the increase in effective nuclear charge. So, this property does increase from iodine to fluorine.

(d) **Oxidizing power:** The oxidizing power of halogens increases as we move from iodine to fluorine. Fluorine is the strongest oxidizing agent among the halogens. So, this property does increase from iodine to fluorine.

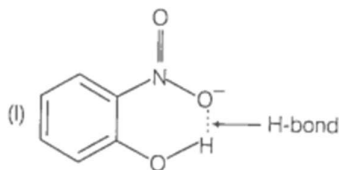
Therefore, the correct combination of properties that increase as we move from iodine to fluorine is:

2. b, c, and d

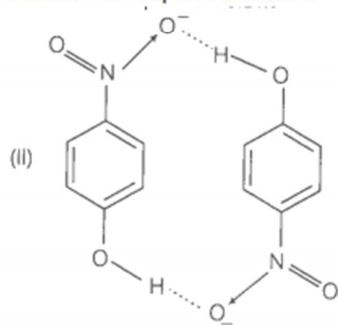
Q59) **Explanation:**

Hint: In ortho-nitrophenol, intramolecular H-bonding occur

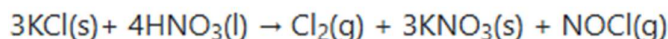
Compound (I) will form intramolecular H-bonding. Intramolecular H-bonding is formed when H-atom, in between the two highly electronegative atoms, is present within the same molecule. In ortho-nitrophenol (compound I), H-atom is in between the two oxygen atoms.



Compound (II) forms intermolecular H-bonding. In para-nitrophenol (II) there is a gap between NO_2 and OH group. So, H-bond exists between H-atom of one molecule and O-atom of another molecule as depicted below.



Q60)



$$\frac{4}{3} \left[\frac{110}{101} \right] \text{ mole} \quad \left(\frac{110}{101} \right) \text{ mole}$$

$$\begin{aligned} \text{Mass of HNO}_3 \text{ used} &= \left[\frac{4}{3} \times \frac{110}{101} \right] \times 63\text{g} \\ &= 91.485\text{g} \end{aligned}$$

Q61) Explanation:

Hint: Sulphur accommodates negative charge more effectively than oxygen

The electron gain enthalpy value for oxygen and sulphur is as follows:

$$\Delta H_{eg}(\text{Oxygen}) = -141 \text{ kJ/mol}$$

$$\Delta H_{eg}(\text{Sulphur}) = -200 \text{ kJ/mol}$$

- **Assertion (A):** "Electron gain enthalpy of oxygen is more than that of sulfur." is **False**.
- **Reason (R):** "Oxygen is more electronegative than sulfur." is **True**.

Therefore, the correct answer is:

3. (A) is False but (R) is True.

Q62) Ionization enthalpy of an element depends on the effective nuclear charge. As effective nuclear charge increases ionization energy also increases.

Ionization enthalpy of an element depends on atomic radius. As effective radius increases the ionization energy decreases.

Half filled and completely filled sub shell has high ionization energy than partially filled subshell. Ionization enthalpy does not depend on the atomic number.

Q63) **Hint:** The stoichiometric ratio of H_2 and H_2O is 1 : 1

Explanation:

Step 1: The reaction for the formation of water is given as follows: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

The given volume of H_2 is 11.2 L

At STP, one mole of any gas will have the volume 22.4 L

Step 2: So, the number of moles of H_2 is $\frac{11.2 \text{ L}}{22.4 \text{ L}} = 0.5 \text{ mol}$

From the balanced equation, the stoichiometric ratio of H_2 and H_2O is 1 : 1

Hence, 0.5 mol of H_2 will form 0.5 mol of H_2O

The mass of water formed = $0.5 \text{ mol} \times 18 \text{ g mol}^{-1} = 9 \text{ g}$

Thus, **option 2** is the correct answer.

Q64) **Explanation:**

Hint: Use uncertainty principle formula

Step 1:

The uncertainty in the speed is 2%. The speed of golf ball is 45 ms^{-1} . Calculate the uncertainty in speed as follows:

$$= 45 \times \frac{2}{100}$$

$$= 0.9 \text{ ms}^{-1}$$

Other given values are:

$$\text{mass of golf ball} = 40 \text{ g} = 40 \times 10^{-3} \text{ kg}$$

$$h = 6.626 \times 10^{-34} \text{ J s}$$

Step 2:

Calculate the value of Δx as follows:

$$\Delta x = \frac{h}{4\pi m \Delta v}$$

$$= \frac{6.626 \times 10^{-34} \text{ Js}}{4 \times 3.14 \times 40 \text{ g} \times 10^{-3} \text{ Kg} \text{ g}^{-1} (0.9 \text{ m s}^{-1})}$$

$$= 1.46 \times 10^{-33} \text{ m}$$

Q65) Hint: The number of d-electrons in Fe^{2+} is 6

Electronic configuration of $\text{Fe}^{2+} = [\text{Ar}]3d^6 4s^0$

$\therefore$ Number of d electrons = 6

Mg - $1s^2 2s^2 2p^6 3s^2$ (6 s electrons)

Cl - $1s^2 2s^2 2p^6 3s^2 3p^5$ (11 p electrons)

Fe - $[\text{Ar}]3d^6 4s^2$ (6 d electrons)

Ne - $1s^2 2s^2 2p^6$ (6 p electrons)

As Cl has 11 electrons in p subshell,

Q66) **Hint:** The formula for the number of moles is $\frac{\text{amount of substance}}{\text{molar mass of substance}}$

Explanation:

Step 1: The formula of the number of moles is equal to $\frac{\text{amount of substance}}{\text{molar mass of substance}}$.

The mole ratio of H_2 and O_2 gas is 8.

$$\frac{n_{\text{H}_2}}{n_{\text{O}_2}} = 8$$

Step 2: Calculate the ratio of weight is as follows: $\frac{\frac{\text{amount of H}_2}{\text{molar mass of H}_2}}{\frac{\text{weight of O}_2}{\text{molar mass of O}_2}} = 8$

Molar mass of H_2 is 2 g mol^{-1} and molar mass of O_2 is 32 g mol^{-1} .

$$\frac{\frac{\text{amount of H}_2}{2}}{\frac{\text{weight of O}_2}{32}} = 8$$

$$16 \times \frac{\text{weight of H}_2}{\text{weight of O}_2} = 8$$

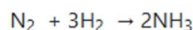
$$\frac{\text{weight of H}_2}{\text{weight of O}_2} = \frac{8}{16} = \frac{1}{2}$$

Hence, **option 4** is the correct answer.

Q67) **Hint:** Follow limiting reagent concept.

Step 1:

The ammonia formation reaction is as follows:



From the reaction, it is clear that 1 mole N_2 reacts with 3 moles H_2 . 1 mole of N_2 is equal to 28 g or 0.028 kg and 3 moles is equal to 6 g or 0.006 kg of H_2 .

0.028 kg N_2 required 0.006 kg of H_2 .

1 kg N_2 required $\frac{0.006}{0.028}$ kg of H_2

2.8 kg required $\frac{0.006}{0.028} \times 2.8$ kg of H_2

= 0.6 kg of H_2

The amount of H_2 needed is 0.6 kg but given amount is 1 kg. Hence, dinitrogen is limiting reagent.

Step 2:

0.028 kg N_2 form 0.034 kg of NH_3 .

1 kg N_2 form $\frac{0.034}{0.028}$ kg of NH_3 .

2.8 kg N_2 form = $\frac{0.034}{0.028} \times 2.8$ kg of NH_3 .

= 3.4 kg of NH_3 or 3400 g

Hence, option third is the correct answer.

Q68) Explanation:

Hint: Electronegativity is not a measurable quantity.

Explanation:

(1) Anionic radii are larger than the parent atom, while cationic radii are smaller because the number of valence electrons has changed while the nuclear charge has remained constant.

An anion has one or more electrons than its parent atom, resulting in increased repulsion among the electrons and a decrease in the effective nuclear charge. As a result, the distance between the valence electrons and the nucleus is greater in anions than in the parent atom.

(2) Electronegativity is the tendency of an atom in a molecule to attract a shared pair of electrons towards itself. As we move down the group, atomic size increases, and due to this, the ability of the nucleus to attract electrons decreases. So, electronegativity decreases in the group.

(3) Electronegativity depends on the force of attraction between the nucleus and bonding electrons in the outer shell.

The electronegativity of any given element is not constant. It varies according to the element to which it is bound. It is not a measurable quantity.

Only statement (I) is correct. Thus, **option 1** is the correct answer.

Q69) Correct Option is b

Q70) Explanation:

Hint: $S.N = \frac{1}{2}[V + M - C + A]$

Calculate the steric number as follows:

$$S.N = \frac{1}{2}[V + M - C + A]$$

Calculate the steric number of the central atom as follows:

$$S.N [\text{Xe}] \text{ in } \text{XeF}_4 = \frac{1}{2}[8 + 4]$$

= 6; that is sp^3d^2

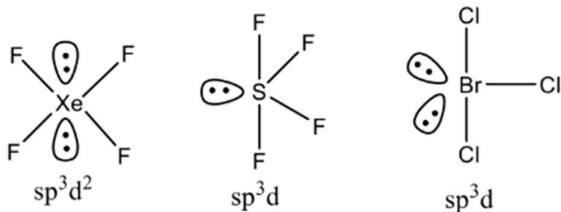
$$S.N [\text{S}] \text{ in } \text{SF}_4 = \frac{1}{2}[6 + 4]$$

= 5; that is sp^3d

$$S.N [\text{Br}] \text{ in } \text{BrCl}_3 = \frac{1}{2}[7 + 3]$$

= 5; that is sp^3d .

The structure is as follows:



Thus, option third is the correct answer.

Q71) Hint: $m_e v r = n \cdot \frac{h}{2\pi}$ $n = 1, 2, 3, \dots$

Step 1:

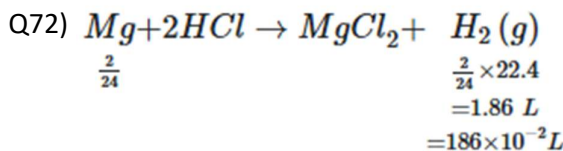
The formula of angular momentum is as follows:

$$m_e v r = n \cdot \frac{h}{2\pi} \quad n = 1, 2, 3, \dots$$

Step 2:

For the 5th orbit, $n=5$

$$\begin{aligned} \text{Angular momentum} &= 5 \cdot \frac{h}{2\pi} \\ &= \frac{2.5h}{\pi} \end{aligned}$$



Q73) **Hint:** Follow Fajan's rule

Explanation:

(1) In Fajan's rule, we find that the chemical bond is covalent or ionic based on the size of the cation or anion. Ionic bonds will have a low positive charge and large cations. Covalent bonds have a high positive charge and small cations.

The ionic character of $CsCl$ is more than $NaCl$. The charge is the same, but the size of the cation is different. Cesium (Cs) has a larger ionic radius than sodium (Na). A higher ionic radius means less polarizing power and greater ionic character.

(2) Lattice energy is directly proportional to the charge on an ion and inversely proportional to the radius of the atom. The radius of Cs^+ is larger than the radius of Na^+ , so the lattice energy of NaCl is higher.

Both assertion and reason are correct, but reason is not the correct explanation of assertion. Hence, **option 2** is the correct answer.

Q74)

$$\begin{aligned} \frac{1}{\lambda} &= R(1)^2 \left(\frac{1}{9} - \frac{1}{\infty} \right) \\ \frac{1}{\lambda} &= R \left(\frac{1}{9} \right) \\ \lambda &= \frac{9}{R} \end{aligned}$$

- Q75) Hint: If central atom orbital does not form hybridized orbital then bond angle is approx 90°

The bond angle of a central atom totally depends on the hybridization state of the central atom. Calculate the hybridization of SO_2 , OH_2 , and NH_3 as follows:

1. SO_2

$$\begin{aligned} S.N &= \frac{1}{2}(V + M - C + A) \\ &= \frac{1}{2}(6 + 0 - 0 + 0) \\ &= 3 \end{aligned}$$

The hybridization is sp^2 bond angle is 119°

2. H_2O

$$\begin{aligned} S.N &= \frac{1}{2}(V + M - C + A) \\ &= \frac{1}{2}(6 + 2 - 0 + 0) \\ &= 4 \end{aligned}$$

The hybridization is sp^3 and due to lone pair-lone pair repulsion the bond angle is less than 109° , it is, 105° .

3. NH_3

$$\begin{aligned} S.N &= \frac{1}{2}(V + M - C + A) \\ &= \frac{1}{2}(5 + 3 - 0 + 0) \\ &= 4 \end{aligned}$$

The hybridization is sp^3 and due to lone pair-bond pair repulsion the bond angle is less than 109° , it is, 107° .

4. In H_2S , hybridization does not take place and pure p orbital participate in bond formation. Hence, bond angle is approx 90° .

The correct option is third.

- Q76) **Explanation:** Bohr model is only valid for $1e^-$ system. So it is not valid for Helium and lithium because they are two electron systems, and others are the isotopes of Hydrogen and they all contains 1 electron only.

Q77) $[\text{Ne}]3s^23p^3$ has highest ionisation energy due to presence of stable configuration

Q78) **Hint:** $E = N_A \times h\nu$

Explanation:

The given frequency (ν) is: $2 \times 10^{14} \text{ Hz}$

The value of Planck's constant (h) is $6.63 \times 10^{-34} \text{ Js}$

Hence, the energy of single photon will be $h\nu$

The energy of one photon will be: $6.63 \times 10^{-34} \text{ Js} \times 2 \times 10^{14} \text{ Hz}$

The energy of one mole of photon is: $6.023 \times 10^{23} \text{ mol}^{-1} \times 6.63 \times 10^{-34} \text{ Js} \times 2 \times 10^{14} \text{ Hz}$

or, the energy of one mole of photon is: $79.8 \times 10^3 \text{ Jmol}^{-1} = 79.8 \text{ kJmol}^{-1}$

Option 3 is the correct answer.

Q79) **Explanation:**

Hint: The masses of oxygen that combines with a fixed mass of hydrogen are in a ratio of small whole number ratio.

For H_2O_2 :

Mass of hydrogen = 5.93 gm

Mass of oxygen = $100 - 5.93 = 94.07 \text{ gm}$

Ratio of mass of hydrogen to mass of oxygen,

i.e., $\text{H}:\text{O} = 5.93 : 94.07$

$\text{H}:\text{O} = 1:16$

For H_2O :

Mass of hydrogen = 11.2 gm

Mass of oxygen = $100 - 11.2 = 88.8 \text{ gm}$

Ratio of mass of hydrogen to mass of oxygen,

i.e., $\text{H}:\text{O} = 11.2 : 88.8$

$\text{H}:\text{O} = 1:32$

Now, As the ratio of mass of hydrogen remains same in both the cases and the ratio of masses of oxygen in both the cases (16:32 or 1:2) is a whole number ratio.

Therefore, the above data illustrates the law of multiple proportions

Q80) Explanation:

Hint: An atom obeys the octet rule when it contains eight electrons (an octet) in its outer shell.

The PF_5 has 10 electrons.

The SF_6 has 12 electrons.

The CCl_4 has 8 electrons.

The BF_3 has 6 electrons.

So, octet is present in the compound, CCl_4 around the central atom

Q81) Explanation:

Hint: A half-filled and fully-filled subshell is more stable than a partially-filled.

Explanation:

The correct electronic configurations of the given elements are as follows:

S.No.	Atomic number	Element	Electronic configuration
1	19	K	$[\text{Ar}]4s^1$
2	46	Pd	$[\text{Kr}]4d^{10}$
3	24	Cr	$[\text{Ar}]3d^5, 4s^1$
4	29	Cu	$[\text{Ar}]3d^{10}, 4s^1$

Chromium is a 3d-series element whose atomic number is 24. In this element, the electron does not fill into a 4s-orbital; instead, it fills into a 3d-orbital, as a half-filled orbital is more stable. So, the electronic configuration of chromium is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$.

The electronic configuration of palladium is $[\text{Kr}]4d^{10}$ and not $[\text{Kr}]4d^8, 5s^2$ because of the stability of a fully-filled electronic configuration.

So, the correct answer is **option 2**.

Q82) STEP1- The metallic character of elements decreases from left to right across a period. Thus, the metallic character of Mg is more than that of Al.

The metallic character of elements increases down a group. Thus, the metallic character of Al is more than that of B.

STEP2- Considering the above statements, we get $K > \text{Mg}$.

Hence, the correct order of metallic character is $K > \text{Mg} > \text{Al} > \text{B}$.

Q83) **Explanation:**

Hint: For octahedral and tetrahedral, the coordination number is 6 and 4 respectively.

The hybridization for the tetrahedral molecule is sp^3 and the coordination number is 4.

The hybridization for the square planar molecule is dsp^2 and the coordination number is 4.

The hybridization for the trigonal bipyramidal molecule is sp^3d and the coordination number is 5.

The hybridization for the octahedral molecule is sp^3d^2 and the coordination number is 6.

Q84) XeF_2 and ICl_2^- are linear.

Q85) The Rydberg formula is given as shown below.

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_i^2} - \frac{1}{n_h^2} \right)$$

Where,

$$\Rightarrow \frac{1}{\lambda'} = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) 2^2$$

$$\Rightarrow \frac{1}{\lambda'} = \frac{4}{A} \left(\frac{1}{9} - \frac{1}{16} \right)$$

$$\Rightarrow \frac{1}{\lambda'} = \frac{4}{A} \times \frac{7}{144}$$

$$\Rightarrow \lambda' = \frac{144A}{28}$$

$$\Rightarrow \lambda' = \frac{36A}{7}$$

Hence, the correct option is the 4th option.

Q86) **Hint:** Isoelectronic species contain the same number of electrons.

Explanation:

In the first case,

$BO_3^{3-} = 32$ electrons, $CO_3^{2-} = 32$ electrons, $NO_3^- = 32$ electrons

In the second case,

$SO_3^{2-} = 42$ electrons, $CO_3^{2-} = 32$ electrons, $NO_3^- = 32$ electrons

In the third case,

$CN^- = 14$ electrons, $N_2 = 14$ electrons, $C_2^{2-} = 14$ electrons

In the fourth case,

$PO_4^{3-} = 50$ electrons, $SO_4^{2-} = 50$ electrons, $ClO_4^- = 50$ electrons

Thus, option 2nd is the correct answer because it does not contain isoelectronic species.

Q87) Explanation:

Hint: One atom of ^{14}C contains 8 neutrons.

Explanation:

Step 1: The given mass of ^{14}C is $7\text{ mg} = 7 \times 10^{-3}\text{ g}$

The aim of this question is to calculate the number of neutrons in the given amount of ^{14}C

The molar mass of $^{14}\text{C} = 14\text{ g mol}^{-1}$

So, the number of moles = $\frac{7 \times 10^{-3}\text{ g}}{14\text{ g mol}^{-1}} = 5 \times 10^{-4}\text{ mol}$

Step 2: The number of atoms of $^{14}\text{C} = 6.023 \times 10^{23} \frac{\text{atoms}}{\text{mol}} \times 5 \times 10^{-4}\text{ mol}$

The number of atoms of $^{14}\text{C} = 30.115 \times 10^{19}\text{ atoms}$

Since one atom of ^{14}C contains 8 neutrons, the total number of neutrons will be:

$$30.115 \times 10^{19}\text{ atoms} \times 8 \frac{\text{neutrons}}{\text{atom}} = 2.409 \times 10^{21}\text{ neutrons}$$

or, the total number of neutrons will be: $24.09 \times 10^{20}\text{ neutrons}$

Hence, **option 2** is the correct answer.

Q88) Explanation:

Hint: From Einstein's Photoelectric Equation

Explanation:

Step 1: The wavelength of initial radiation: $250\text{ nm} = 2.5 \times 10^{-7}\text{ m}$

The energy of incident radiation = $\frac{h c}{\lambda}$

$$\text{The energy of incident radiation} = \frac{6.63 \times 10^{-34}\text{ J s} \times 3 \times 10^8\text{ m s}^{-1}}{2.5 \times 10^{-7}\text{ m}} = 7.96 \times 10^{-19}\text{ J}$$

We also know that, $1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$

So, the energy of incident radiation = 4.97 eV

Step 2: The potential applied gives the kinetic energy of the electron.

So, the kinetic energy of the electron = 0.5 eV

$$\text{The work function} = (4.96 - 0.5)\text{ eV} = 4.46\text{ eV}$$

The work function of the metal is 4.5 eV . Hence, **option 3** is the correct answer.

Q89) **Explanation:**

Hint: Penetration effect can be used to determine the attraction of electrons in the orbitals by the nucleus. The order is: $s > p > d > f$

Q90) F^- & O^- has zero electron affinity because no energy released in the addition of electron.
In O & Na, O has highest electron affinity



Mitoprep

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